ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 27-Sep-2026

Site: A | Location: Shivamogga | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
1.25 kW	**1,982 kWh**	**Rs. 102,075**	**7Y 2M**
1.65 kWp Installed	CUF 18.1% | PR 75%	Rs. 1.02 Lakhs	Profit: Rs. 5.33L

[SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

Company / Customer Name	A
Contact Mobile / Phone	Not Specified
Location / Address	Shivamogga
Contact Email Address	voltroof4@gmail.com
Design Path Selected	Monthly Electricity Bill Units
System Target Capacity	1.25 kW (Single LT Installation)
Total Installed Array Rating	1.65 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	3 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	125.00 Sq.Feet (11.61 sq.m)
Utilized Usable Rooftop Area	125 Sq.Feet (11.61 sq.m)
First Year Annual Generation	1,981.91 kWh
25-Year Cumulative Generation	44,033.82 kWh
Estimated Payback Period	7 Years & 2 Months
Post-Payback Profitability (25 Yrs)	Rs. 533,036.39 (INR 5.33 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 492,787.39 (INR 4.93 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 36 Tonnes CO2 | Tree Equivalence: 57 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Building 1 (Rooftop Array)
Gross Rooftop Area	125.00 Sq.Feet (11.61 sq.m)
Utilized Effective Rooftop Area	125 Sq.Feet (11.61 sq.m)
Building Plant Target Capacity	1.25 kW
Solar PV Module Rating	550 W Mono-PERC (Mono-facial)
Number of Solar Modules	3 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	2 kW, 230V 1-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (2 kW)

Core inverter rating (100% direct sizing), MPPT topology & models

Selected Inverter Rating	2 kW Grid-Tied Inverter (Single-Phase)
Certified Top-Tier OEM Models	Polycab PSIT, Obetech G
Direct AC Design Power (100% Sizing)	1.25 kW AC (from 1.25 kWp DC)
AC Output Voltage & Grid Phase	230V, 1-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx4 Sq.mm
DC Array Isolator Switch	16A, MCB Type, 2P 230V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	16 Amps
Rated Inverter Operating Current	5.43 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx4 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	25A, 230V, 2PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	100.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV Modules (Mono PERC / TC	550W Mono PERC Half-Cut Cells (>21.3% Efficiency)	3 Nos.	Rs. 30,000
02	On-Grid Solar String Inverter(s)	Dual MPPT, IP65 (A: 2 kW)	1 Set(s)	Rs. 36,000
03	Mounting structure	HDGI rooftop structure	1 Site(s)	Rs. 6,000
04	DC solar cable	4 sq.mm copper	1 Site(s)	Rs. 1,400
05	AC cable	4 sq.mm copper	1 Site(s)	Rs. 1,275
06	MC4 connectors	PV-rated connector pairs	1 Site(s)	Rs. 900
07	DCDB	Isolator + fuse + DC SPD	1 Site(s)	Rs. 3,500
08	ACDB	16A MCB + 25A RCCB + Type-II AC SPD	1 Site(s)	Rs. 3,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	1 Site(s)	Rs. 4,500
10	Lightning arrester	Standard	1 Site(s)	Rs. 2,500
11	Installation accessories	Lugs, ferrules, clamps, conduit, etc.	1 Site(s)	Rs. 2,500
12	Installation & commissioning	Labour and commissioning	1 Site(s)	Rs. 10,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 102,075.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 500 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 1,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 4,500 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 3,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 102,075 | ESCOM Utility Budget: Rs. 9,012 | Grand Total Outlay: Rs. 111,087 (Rs. 1.11 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 102,075	**7Y 2M**	**Rs. 5.33L**	**Rs. 4.93L**
12-Item BOM Basis	Break-Even Year	Rs. 533,036	Alternative Asset

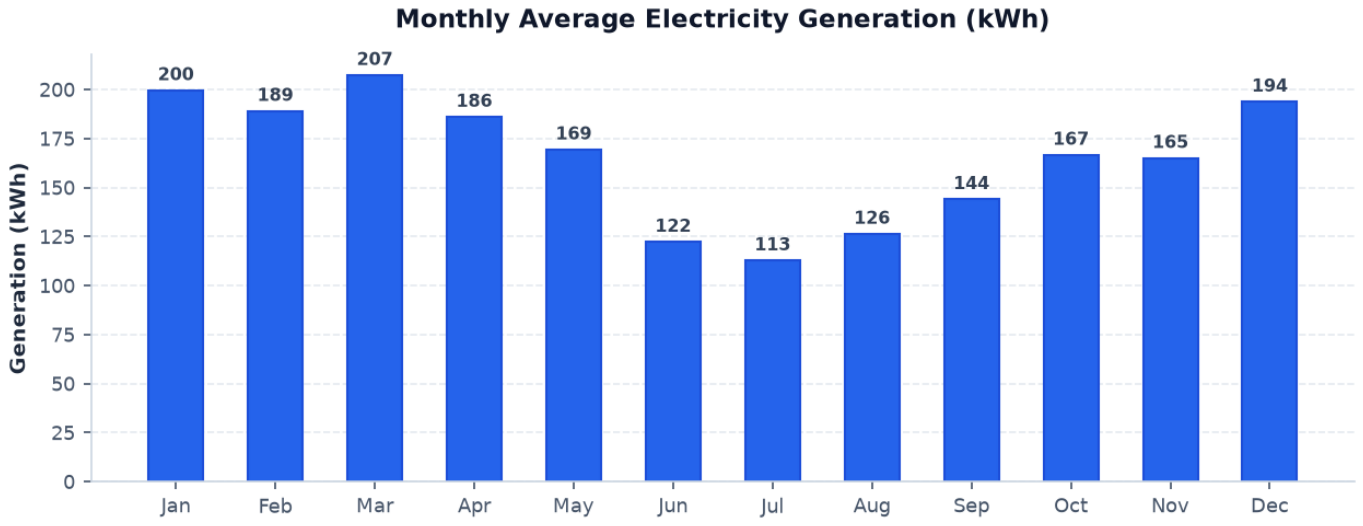
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	1,982	Rs. 8.05	15,954	750	15,204	15,204
02	1,962	Rs. 8.45	16,580	750	15,830	31,034
03	1,942	Rs. 8.87	17,230	750	16,480	47,514
04	1,923	Rs. 9.32	17,923	750	17,173	64,687
05	1,904	Rs. 9.78	18,619	750	17,869	82,556
06	1,885	Rs. 10.27	19,357	750	18,607	101,163
07 *	**1,866**	**Rs. 10.78**	**20,115**	**750**	**19,365**	**120,527**
08	1,847	Rs. 11.32	20,911	750	20,161	140,688
09	1,829	Rs. 11.89	21,744	750	20,994	161,683
10	1,811	Rs. 12.48	22,595	750	21,845	183,528
11	1,792	Rs. 13.11	23,498	750	22,748	206,276
12	1,774	Rs. 13.76	24,417	750	23,667	229,943
13	1,757	Rs. 14.45	25,385	750	24,635	254,578
14	1,739	Rs. 15.18	26,401	750	25,651	280,228
15	1,722	Rs. 15.93	27,428	750	26,678	306,906
16	1,705	Rs. 16.74	28,534	750	27,784	334,691
17	1,688	Rs. 17.57	29,650	750	28,900	363,590
18	1,671	Rs. 18.45	30,823	750	30,073	393,664
19	1,654	Rs. 19.37	32,037	750	31,287	424,950
20	1,637	Rs. 20.34	33,305	750	32,555	457,505
21	1,621	Rs. 21.36	34,625	750	33,875	491,380
22	1,605	Rs. 22.43	35,996	750	35,246	526,626
23	1,589	Rs. 23.55	37,415	750	36,665	563,291
24	1,573	Rs. 24.73	38,897	750	38,147	601,438
25	1,557	Rs. 25.96	40,423	750	39,673	641,111

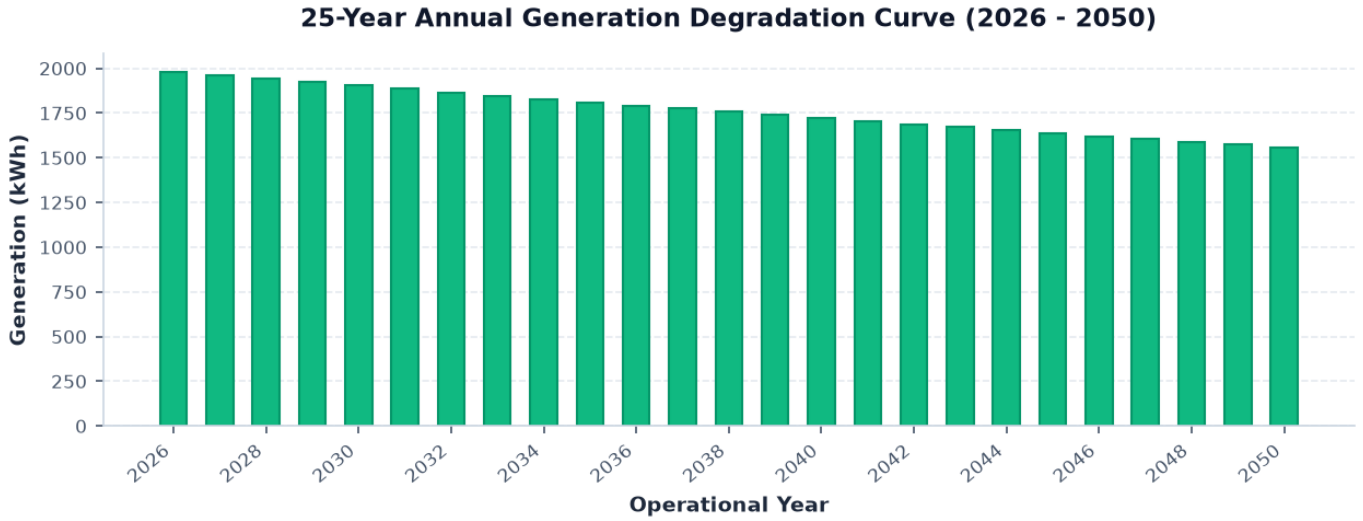
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1586 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

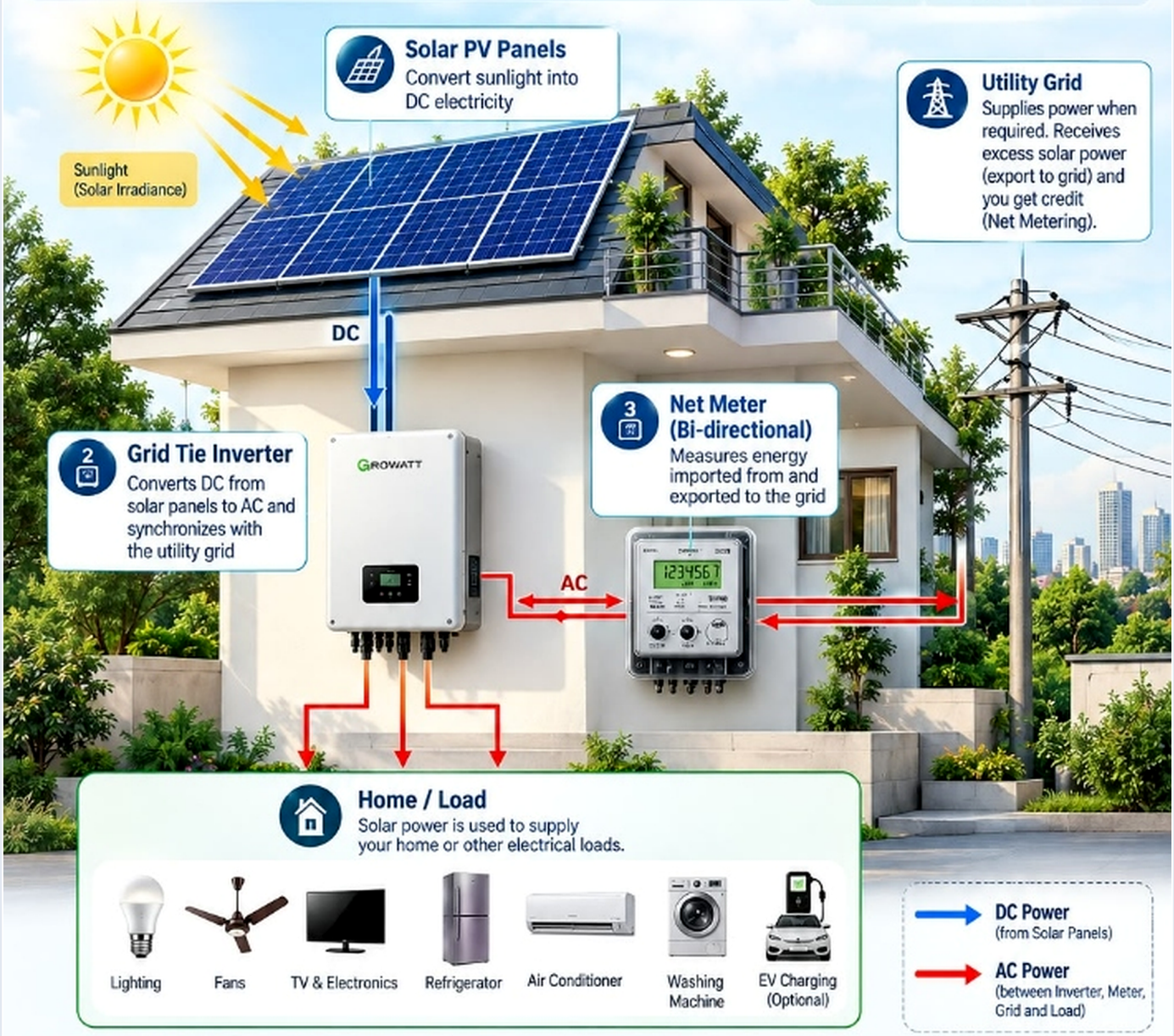
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow